INCH-POUND
MIL-C-48485C(AR)
23 June 1995
(see 6.4)

MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY - COMPARATOR, TYPE 3

This specification is approved for use by the U.S. Army Armament, Research, Development and Engineering Center (ARDEC) is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements and quality assurance provisions for the Circuit Card Assembly - Comparator Type 3, 11732535 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-13926 - Fire Control Materiel, General Specification Governing the Manufacture and Inspection of

MIL-I-45607 - Inspection Equipment; Acquisition, Maintenance, and Disposition of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use to improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: AMSTA-AR-EDE-S, Picatinny Arsenal, New Jersey 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1240

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STANDARDS

MILITARY

MIL-STD-100	- Engineering Drawing Practices
MIL-STD-454	- Standard General Requirements for Electronic Equipment
MIL-STD-810	- Environmental Test Methods
MIL-STD-2000	- Standard Requirements for Soldered Electrical and Electronic Assemblies
MIL-STD-2073-1	- Procedures for Development and Application of Packaging Requirements

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from: DODSSP - Customer Service, Standardization Documents Order Desk, 700 Robbins Avenue, Bldg 4D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issue shall be those in effect on date of the solicitation.

U.S. Army Armament Research Development and Engineering Center (ARDEC)

DRAWINGS

11732535	- Circuit Card Assembny Comparator, Type 3

(Copies of Government drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets or MS standards), the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtainem (see contract provisions for additional precedence criteria).

3. REQUIREMENTS

3.1 Fabrication. The comparator, type 3 circuit card assembly, herein referred to as the assembly, shall be manufactured in accordance with Drawing 11732535 and drawings pertaining thereto and, when assembled, shall meet the requirements of this specification (see 4.5.1).

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3.1.1 Function. The assembly shall provide the following:

a. Compare the ($W\text{-}K_WV_W$) T_F and V_W signals with predetermined reference signals.

b. Decode the T_G FAIL and $\overline{\text{WS FAIL}}$ signals from the computer.

c. Decode comparator outputs and other logic signals from the Field Test Set to allow either COMPUTER OK or CARD MALFUNCTION indications.

3.1.2 General specifications. (see 4.6.3)

3.1.2.1 Manufacture and inspection. The following provisions of MIL-F-13926 apply:

a. Dimensions and tolerances
b. Effect of protective coating on dimensions (Inorganic type coatings)
c. Part identification and marking
d. Workmanship (including applicable portions of MIL-STD-2000 and MIL-STD-454, Requirement 9)

3.1.2.2 Standards of manufacture.

3.1.2.2.1 Assembly and soldering. The requirements of MIL-STD-2000 shall apply, as a minimum.

3.1.2.2.2 Interchangeability. The assembly shall be manufactured in accordance with the interchangeability requirements as specified in MIL-STD-100.

3.1.3 Ambient conditions. Standard ambient conditions shall be as follows:

a.	Temperature	73° ± 18°F
b.	Relative humidity	50 percent ± 30 percent
c.	Atmospheric pressure	28.5 + 2.0 -3.0 in. Hg.

3.2 First article. When specified in the contract or purchase order (see 6.2), the contractor shall furnish sample units for first article inspection and approval (see 4.4).

3.3 Performance. Unless otherwise specified, the assembly shall meet the performance requirements specified herein under standard ambient conditions of 3.1.3.

3.3.1 ($W\text{-}K_WV_W$) T_F comparator logic. With the power sources (items 2.2 thru 2.6 of Table I), signal sources (items 3.1, 3.2, and 3.3 of Table I) and interconnection (item 4.1 of Table I) applied to the assembly; and with each item's input conditions

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of Tables II and III established in any corresponding item number order (output at pin P1-A17 with respect to pin P1-A13) and: (see 4.6.1.1)

a. With the inputs applied as specified by the Table II item, and with the variable V rms signal adjusted to the nominal center point reading specified by the corresponding Table III item, the output shall be 5.0 +0.5V dc/-2.6V dc.

b. With the variable V rms signal decreased from nominal, the output shall change to 0.0 ±0.5V dc. This change shall occur with the variable V rms signal adjusted within the outward bound lower limit specified in Table III item.

c. With the variable V rms signal increased from the outward bound lower limit, the output shall change to 5.0 +0.5V dc/ -2.6V dc. This change shall occur with the variable V rms signal adjusted within the inward bound lower limit specified in the Table III item.

d. With the variable V rms signal increased from the inward lower limit, the output shall change to 0.0 ± 0.5V dc. This change shall occur with the variable V rms signal adjusted within the outward bound upper limit specified in the Table III item.

e. With the variable V rms signal decreased from the outward upper limit, the output shall change to 5.0 +0.5V dc/ -2.6V dc. This change shall occur with the variable V rms signal adjusted within the inward bound upper limit specified in the Table III item.

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TABLE I. Loads, power, signals and interconnections.

Item	Condition	Characteristics	Connections
1.0	Loads		Connected between the following pins:
1.1	Lamp	24V lamp in series with 24V dc	P1-B11(-) and 24V dc(+) (See item 2.1)
1.2	Lamp	24V lamp in series with 24V dc	P1-B16(-) and 24V dc(+) (See item 2.1)
1.3	Lamp	24V lamp in series with 24V dc	P1-A16(-) and 24V dc(+) (See item 2.1)
1.4	Lamp	24V lamp in series with 24V dc	P1-A22(-) and 24V dc(+) (See item 2.1)
1.5	Lamp	24V lamp in series with 24V dc	P1-B29(-) and 24V dc(+) (See item 2.1)
1.6	Lamp	24V lamp in series with 24V dc	P1-A29(-) and 24V dc(+) (See item 2.1)
2.0	Power Sources		Connected between the following pins:
2.1	24V dc	Tolerance: ±2V dc Ripple: 200 mV p-p max Current: 500 ma min	P1-A23(-) and load lamps (items 1.1 through 1.6)
2.2	15.00V dc	Tolerance: ±0.03V dc Ripple: 20 mV p-p max Current: 500 ma min	P1-A1(+) and P1-A3(-)
2.3	-15.00V dc	Tolerance: ±0.03V dc Ripple: 20 mV p-p max Current: 500 ma min	P1-A4(-) and P1-A3(+)
2.4	5.25V dc	Tolerance: ±0.15V dc Ripple: 20 mV p-p max Current: 500 ma min	P1-A30(+) and P1-A13(-)

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TABLE I. Loads, power, signals and interconnections. - Continued

Item	Condition	Characteristics	Connections
2.5	10.000V dc	Tolerance: ±0.010V dc Ripple: 20 mV p-p max Current: 500 ma min	P1-A7(+) and P1-A3(-)
2.6	-10.000V dc	Tolerance: ±0.010V dc Ripple: 20 mV p-p max Current: 500 ma min	P1-A10(+) and P1-A3(-)
3.0	Signal Sources		Connected between the following pins:
3.1	30 V, 400 Hz (MR1)	Squarewave: +15V to -15V Tolerances: ±0.50V, ±1 Hz phase: 1	P1-A6 and P1-A3 (rtn)
3.2	30 V, 400 Hz (MR2)	Squarewave: +15V to -15V Tolerances: ±0.50V, ±1 Hz phase: 2 (180° out-of-phase with item 3.1)	P1-B7 and P1-A3 (rtn)
3.3	4.000V rms, 400 Hz	Sinewave Tolerances: ±0.002V rms, +1 Hz phase: 2	P1-B3 and P1-A3 (rtn)
3.4	variable V rms, 400 Hz	Sinewave: 0.330 to 2.450V rms, variable Tolerances: ±1 Hz phase: 1	P1-B1 and P1-A3 (rtn)
3.5	variable V dc (wind velocity)	dc level: -4.370 to +4.370V dc, variable	P1-A8 and P1-A3 (rtn)
3.6	Test Enable Clock	See Figure 1.	P1-B27 and P1-A13 (rtn)
3.7	Test In Progress Clock	See Figure 1.	P1-A27 and P1-A13 (rtn)

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TABLE I. Loads, power, signals and interconnections. - Continued

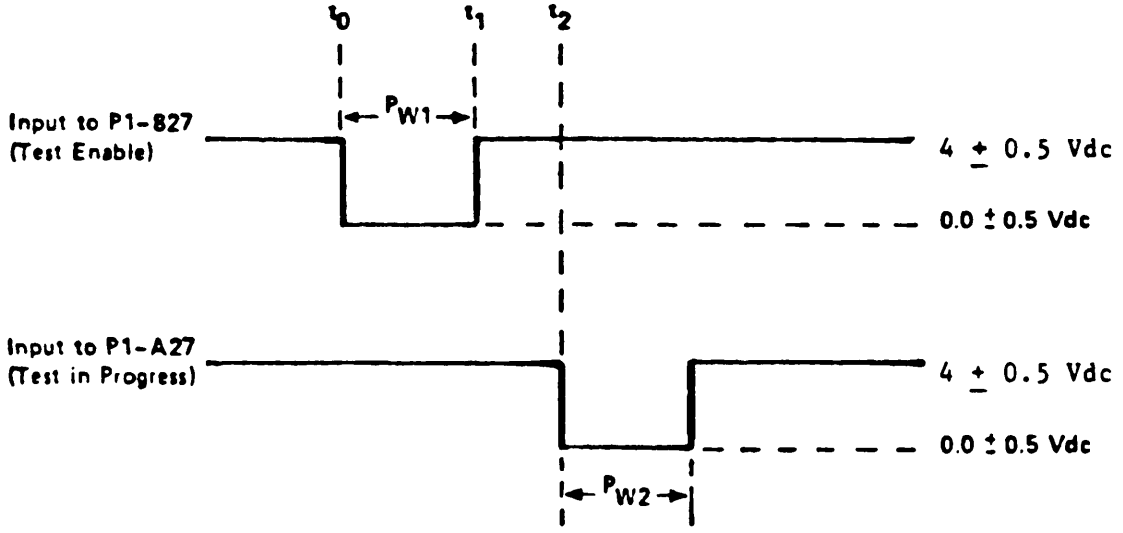
Item	Condition	Characteristics	Connections
3.8	Digital Signal FET Control No. 1	"0" level: -15.00 ± 0.030V dc "1" level: 0.0 ± 0.5V dc Ripple: 20 mvp-p max Current: 300 ma min	Applied as specified with return to P1-A13
3.9	Digital Signal FET Control No. 2	"0" level: -15.00 ± 0.030V dc "1" level: +15.000 ± 0.030V dc Ripple: 20 mvp-p max Current: 300 ma min	Applied as specified with return to P1-A3
3.10	Digital Signal logic control	"0" level: 0.0 ± 0.5V dc "1" level: 5.0 +0.5V dc/-2.6V dc Ripple: 20 mvp-p max Current: 300 ma min	Applied as specified with return to P1-A13
4.0	Interconnections		Interconnection of the following pins:
4.1			P1-A3 to P1-A13

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TABLE II. $(W-K_W V_W)T_F$ comparator scaling select input conditions.

Item	Input Conditions at P1 Pins					
	B6 1/	A5 1/	B5 1/	A2 1/	B4 1/	A15 2/
1	1	1	0	0	0	0
2	0	1	0	0	0	0
3	0	0	1	0	0	0
4	0	0	0	1	0	0
5	0	0	0	0	1	0
6	0	0	0	0	1	1

1/ The notations "0" and "1" refer to FET control No. 1 digital signal defined as item 3.8 of Table I.
2/ The notations "0" and "1" refer to the logic control digital signal defined as item 3.10 of Table I.



P_{W1}>25 milliseconds

P_{W2}>20 milliseconds

$t_2 > t_1$+25 milliseconds

FIGURE 1. Input timing diagram at P1-A27 and P1-B27.

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TABLE III. $(W-K_WV_W)T_F$ comparator detection windows. 1/

Item	Lower Limit		Nominal	Upper Limit	
	Outward Bound	Inward Bound	Center Point	Inward Bound	Outward Bound
1	0.342 to 0.358	0.353 to 0.368	0.397 ± 0.005	0.424 to 0.439	0.433 to 0.449
2	0.370 to 0.388	0.382 to 0.398	0.426 ± 0.005	0.452 to 0.468	0.461 to 0.478
3	0.847 to 0.867	0.859 to 0.877	0.904 ± 0.005	0.929 to 0.948	0.938 to 0.958
4	0.331 to 0.348	0.342 to 0.358	0.387 ± 0.005	0.413 to 0.429	0.422 to 0.439
5	2.158 to 2.183	2.170 to 2.193	2.218 ± 0.005	2.240 to 2.264	2.250 to 2.274
6	2.105 to 2.142	2.128 to 2.163	2.218 ± 0.005	2.269 to 2.304	2.287 to 2.324

1/ All readings in V rms and refer to the variable V rms signal defined as item 3.4 of Table I.

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3.3.2 V_W comparator logic. With the power sources (items 2.2 thru 2.6 of Table I), signal sources (items 3.1 and 3.2 of Table I), and interconnection (item 4.1 of Table I) applied to the assembly; and with each item's input conditions of Table IV established in any item number order, with the output at pin P1-B18 measured with respect to pin P1-A13 and: (see 4.6.1.2)

a. With the input applied as specified by the Table IV item, and with the variable V dc signal adjusted to the nominal center point reading specified by the corresponding Table IV item, the output shall be 0.0 ± 0.5V dc.

b. With the variable signal decreased from nominal, the output shall change to 5.0 +0.5/-2.6V dc. This change shall occur with the variable V dc signal adjusted within the outward bound lower limit specified in the Table IV item.

c. With the variable V dc signal increased from the outward lower limit, the output shall change to 0.0 ± 0.5V dc. This change shall occur with the variable V dc signal adjusted within the inward bound lower limit specified in the Table IV item.

d. With the variable V dc signal increased from the inward bound lower limit, the output shall change to 5.0 +0.5/ -2.6V dc. This change shall occur with the variable V dc signal adjusted within the outward bound upper limit specified in the Table IV item.

e. With the variable V dc signal decreased from the outward bound upper limit, the output shall change to 0.0 ± 0.5V dc. This change shall occur with the variable V dc signal adjusted within the inward bound upper limit specified in the Table IV item.

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TABLE IV. V_W comparator input and detection windows.

	INPUTS 1/	DETECTION WINDOWS 2/				
		Lower Limit			Upper Limit	
Item	Applied to Pin P1-B23	Outward Bound	Inward Bound	Nominal Center Point	Inward Bound	Outward Bound
1	0	3.783 ± 0.033	3.842 ± 0.032	4.020 ± 0.005	4.186 ± 0.032	4.239 ± 0.033
2	1	-4.251 ± 0.050	-4.193 ± 0.049	-4.014 ± 0.005	-3.848 ± 0.049	-3.796 ± 0.049

1/ The input notations "0" and "1" refer to FET control No. 2 digital signal defined as item 3.9 of Table I.

2/ Detection window readings are in V dc and refer to the variable V dc (wind velocity) signal defined as item 3.5 of Table I.

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3.3.3 Computer A9 card fail logic. With the power sources (items 2.1 thru 2.6 of Table I), load lamp (item 1.2 of Table I), and interconnection (item 4.1 of Table I) applied to the assembly; and with each item's input conditions of Table V applied in any item number order, the state of the load lamp shall be as specified in Table V (see 4.6.1.3).

TABLE V. Computer A9 card fail logic.

Item	Inputs at P1 Pin 1/, 2/						Lamp 3/ Status
	B10	A25	A14	A12	A20	B12	
1	1	0	0	0	0	0	OFF
2	1	0	0	0	1	0	ON
3	1	0	1	0	1	0	OFF
4	1	0	0	NC	0	0	ON
5	0	1	0	0	0	0	ON
6	0	1	0	0	1	0	OFF
7	0	1	0	NC	0	1	ON

1/ The input notations "0" and "1" refer to the logic control digital signal defined as item 3.10 of Table I.
2/ The input notation "NC" specifies a "no connection" condition.
3/ Lamp status "ON" indicates an illuminated lamp; "OFF" indicates an extinguished lamp.

3.3.4 Computer A4 card fail logic. With the power sources (items 2.1 thru 2.6 of Table I), load lamp (item 1.3 of Table I), signal sources (items 3.1, 3.2, and 3.3 of Table I), and interconnection (item 4.1 of Table I) applied to the assembly; and: (see 4.6.1.4)

a. With the "0" level logic control digital signal (item 3.10 of Table I) applied to pin P1-B26, and with each item's input conditions of Table VI established in any item number order, the state of the load lamp shall be as specified in Table VI.

b. With the "0" level logic control digital signal (item 3.10 of Table I) applied to pin P1-A21, and with each item's input conditions of Table VII established in ascending item number order, the state of the load lamp shall be as specified in Table VII.

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TABLE VI. Computer A4 card fail logic (not lead lock).

Item	Inputs at P1 Pin 1/						Input Conditions		Lamp Status
	A21	B15	A20	B20	B25	A25	G 2/	F 3/	4/
1	1	1	1	0	1	0	1	1	ON
2	1	0	1	0	1	0	1	1	OFF
3	1	1	1	0	1	1	1	1	OFF
4	1	1	1	0	1	1	0	1	ON
5	1	1	1	0	0	1	0	1	OFF
6	1	1	0	0	0	1	0	1	ON
7	1	1	0	1	1	1	0	0	OFF
8	0	1	0	0	1	1	0	0	OFF

1/ The input notations "0" and "1" refer to the logic control digital signal defined as item 3.10 of Table I.

2/ Establish the input condition of item 1 of Table IV. The input notation "0" requires the variable V dc signal of Table IV to be adjusted to the nominal center point. The input notation "1" requires the variable V dc signal of Table IV to be adjusted to +4.285 ± 0.010V dc.

3/ Establish the input conditions of item 5 of Table II and Table III. The input notation "0" requires the variable V rms signal of Table III to be adjusted to the nominal center point. The input notation "1" requires the variable V rms signal of Table III to be adjusted to +2.295 ± 0.010 V rms.

4/ Lamp status "ON" indicates an illuminated lamp; "OFF" indicates an extinguished lamp.

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TABLE VII. Computer A4 card fail logic (lead lock).

Item	Inputs at P1 Pin 1/ B15	A25	B26	Input Conditions G 2/	F 3/	Pulse 4/	Lamp Status 5/
1	0	0	1	1	0	yes	OFF
2	0	0	1	1	1	yes	ON
3	1	0	1	1	1	no	OFF
4	1	0	1	1	1	yes	OFF
5	0	0	1	1	1	no	OFF
6	0	0	1	1	1	yes	ON
7	0	0	0	1	1	no	OFF
8	0	1	1	1	1	no	OFF

1/ The input notations "0" and "1" refer to the logic control digital signal defined as item 3.10 of Table I.

2/ Establish the input condition of item 1 of Table IV. The input notation "1" requires the variable V dc signal of Table IV to be adjusted to +4.285 ± 0.010V dc.

3/ Establish the input conditions of item 5 of Table II and Table III. The input notation "0" requires the variable V rms signal of Table III to be adjusted to the nominal center point. The input notation "1" requires the variable V rms signal of Table III to be adjusted to +2.295 ± 0.010V rms.

4/ The notation "yes" requires the application of the pulse signals defined as items 3.6 and 3.7 of Table I. The pulse signals are applied after establishing the specified input conditions. The notation "no" indicates that the pulse signals are not be applied.

5/ Lamp status "ON" indicates an illuminated lamp; "OFF" indicates an extinguished lamp.

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3.3.5 Computer A5 card fail logic. With the power sources (items 2.1 thru 2.6 of Table I), load lamp (item 1.4 of Table I), and interconnection (item 4.1 of Table I) applied to the assembly; and with each item's input conditions of Table VIII established in any item number order, the state of the load lamp shall be as specified in Table VIII (see 4.6.1.5).

TABLE VIII. Computer A5 card fail logic.

Item	Inputs at P1 Pin 1/		Lamp Status 2/
	B24	A14	
1	1	0	OFF
2	0	1	OFF
3	0	0	ON
4	1	1	OFF

1/ The input notations "0" and "1" refer to the logic control digital signal defined as item 3.10 of Table I.

2/ Lamp status "ON" indicates an illuminated lamp; "OFF" indicates an extinguished lamp.

3.3.6 Computer A6 card fail logic. With the power sources (items 2.1 thru 2.6 of Table I), load lamp (item 1.5 of Table I), and interconnection (item 4.1 of Table I) applied to the assembly; and with each item's input conditions of Table IX established in any item number order, the state of the load lamp and the corresponding output at pin P1-B21 shall be as specified in Table IX (see 4.6.1.6).

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TABLE IX. Computer A6 card fail logic.

Item	Inputs at P1 Pin 1/								Lamp Status 2/	Output at Pin P1-B21 3/
	B22	B28	A15	A26	B19	B15	A19	A14		
1	0	1	1	0	0	1	1	0	ON	d
2	0	1	1	0	0	1	0	0	OFF	d
3	0	1	1	0	0	1	0	0	OFF	HI
4	0	1	0	0	0	1	1	0	OFF	HI
5	1	1	1	0	0	1	1	0	OFF	d
6	1	0	0	0	0	1	1	0	OFF	d
7	1	0	0	1	0	1	1	0	ON	LO
8	1	0	0	0	1	1	1	0	ON	d
9	1	0	0	0	1	0	1	0	OFF	d

1/ The input notations "0" and "1" refer to the logic control digital signal defined as item 3.10 of Table I.

2/ Lamp status "ON" indicates an illuminated lamp; "OFF" indicates an extinguished lamp.

3/ Output notation "d" indicates no measurement is required; "HI" indicates a level greater than +2.40V dc; "LO" indicates a level of 0.0 ± 0.5V dc. All outputs are with respect to pin P1-A13.

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3.3.7 Computer A7 card fail logic. With the power sources (items 2.1 thru 2.6 of Table I), load lamp (item 1.6 of Table I), and interconnection (item 4.1 of Table I) applied to the assembly; and with each item's input conditions of Table X established in any item number order, the state of the load lamp shall be as specified in Table X (see 4.6.1.7).

TABLE X. Computer A7 card fail logic.

Item	Inputs at P1 Pin 1/			Lamp Status 2/
	B17	A18	A14	
1	1	0	0	ON
2	1	0	1	OFF
3	0	1	0	ON
4	0	0	0	OFF

1/ The input notations "0" and "1" refer to the logic control digital signal as item 3.10 of Table I.

2/ Lamp status "ON" indicates an illuminated lamp; "OFF" indicates an extinguished lamp.

3.3.8 Computer OK logic. With the power sources (all items 2.1 to 2.6 of Table I), load lamp (item 1.1 of Table I), signal sources (items 3.1, 3.2 and 3.3 of Table I), interconnection (item 4.1 of Table I), and static input conditions (see TABLE XI) applied to the assembly; and with each item's input conditions of Table XII established in any item number order, the state of the load lamp shall be as specified in Table XII (see 4.6.1.8).

TABLE XI. Computer OK test static inputs.

Item	Inputs at P1 Pin						Input Condition F 2/
	A25	A15	B17	A20	B20	B22	
1	0	1	0	0	0	0	1

1/ The input notations "0" and "1" refer to the logic control digital signal defined as item 3.10 of Table I.

2/ Establish the input condition of item 5 of Table II and Table III. The input notation "1" requires the variable V rms signal of Table III to be adjusted to +2.295 ± 0.010 V rms.

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TABLE XII. Computer OK logic.

Item	Inputs at P1 Pin												Input Condition Pulse 2/	Lamp Status 3/
	A21	B8	A9	B9	B14	B24	A18	A24	A19	B10	B15	B26		
1	1	1	1	1	0	1	0	1	0	0	0	0	yes	ON
2	0	1	1	1	0	1	0	1	0	0	0	0	no	OFF
3	1	1	1	1	0	1	0	1	0	0	0	1	no	OFF
4	1	0	1	1	0	1	0	1	0	0	1	0	no	OFF
5	1	1	1	1	0	1	0	1	0	0	1	0	no	OFF
6	1	1	0	1	0	1	0	1	0	0	0	0	no	OFF
7	1	1	1	0	0	1	0	1	0	0	0	0	no	OFF
8	1	1	1	1	1	1	0	1	0	0	0	0	no	OFF
9	1	1	1	1	0	0	0	1	0	0	0	0	no	OFF
10	1	1	1	1	0	1	0	1	1	0	0	0	no	OFF
11	1	1	1	1	0	1	1	1	0	0	0	0	no	OFF
12	1	1	1	1	0	1	0	0	0	0	0	0	no	OFF
13	1	1	1	1	0	1	0	1	0	1	0	0	no	OFF

1/ The input notations "0" and "1" refer to the logic control digital signal defined as item 3.10 of Table I.

2/ The notation "yes" requires the application of the pulse signals defined as items 3.6 and 3.7 of Table I. The pulses are applied after establishing the specified input conditions. The notation "no" indicates that the signals are not to be applied.

3/ Lamp status "ON" indicates an illuminated lamp; "OFF" indicates an extinguished lamp.

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3.3.9 Dot D_L output logic. With the power sources (items 2.1 thru 2.6 of Table I), and interconnection (item 4.1 of Table I) applied to the assembly; and with each item's input conditions of Table XIII established in any item number order, the corresponding output at pin P1-B30 shall be as specified in Table XIII (see 4.6.1.9).

TABLE XIII. D_L output logic.

Item	Inputs at P1 Pin 1/			Output at Pin
	A15	A28	B13	P1-B30 2/
1	1	0	1	LO
2	1	1	0	LO
3	1	1	0	HI
4	0	1	0	HI

1/ The input notations "0" and "1" refer to the logic control digital signal defined as item 3.10 of Table I.

2/ The output notation "LO" indicates a level of 0.0 ± 0.5V dc; "HI" indicates a level greater than +2.40V dc. All outputs are with respect to pin P1-A13.

3.4 Environmental.

3.4.1 Temperature. The assembly shall be capable of operating as specified herein over the operating temperature range of +40°F to 100°F and shall not be damaged when subjected to the storage conditions ranging from -65°F to 160°F (see 4.6.2.1 and 4.6.2.2).

3.4.2 Vibration. The assembly shall be capable of operating as specified herein after withstanding exposure under the conditions of 3.1.3 to a sinusoidal vibration displacement of .03 inch double amplitude, 10 - 55 Hz in accordance with Figure 2. Duration of exposure shall be not less than 80 minutes in each of three mutually perpendicular axes (see 4.6.2.3).

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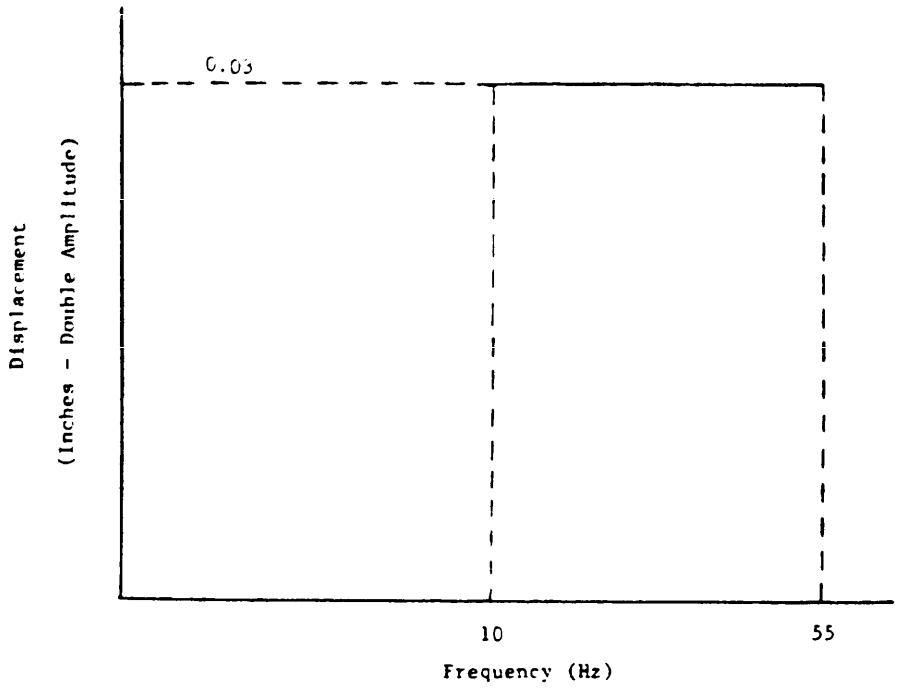


FIGURE 2. Vibration Curve.

3.4.3 Shock. The assembly shall be capable of operating as specified herein after withstanding exposure under the conditions of 3.1.3 to half sine wave shock pulses of 40 ± 4 gravity unit (g's) for a duration of 18 ± 3 milliseconds (msec) in accordance with Figure 3. The assembly shall withstand three shock pulses applied in both directions along three mutually perpendicular axes (18 shocks). In addition, the assembly shall be capable of withstanding three half sine wave shock pulses of 100.0 ± 10.0 g's for a duration of 1.5 ± 0.2 msec applied in each direction along three mutually perpendicular axes (see 4.6.2.4).

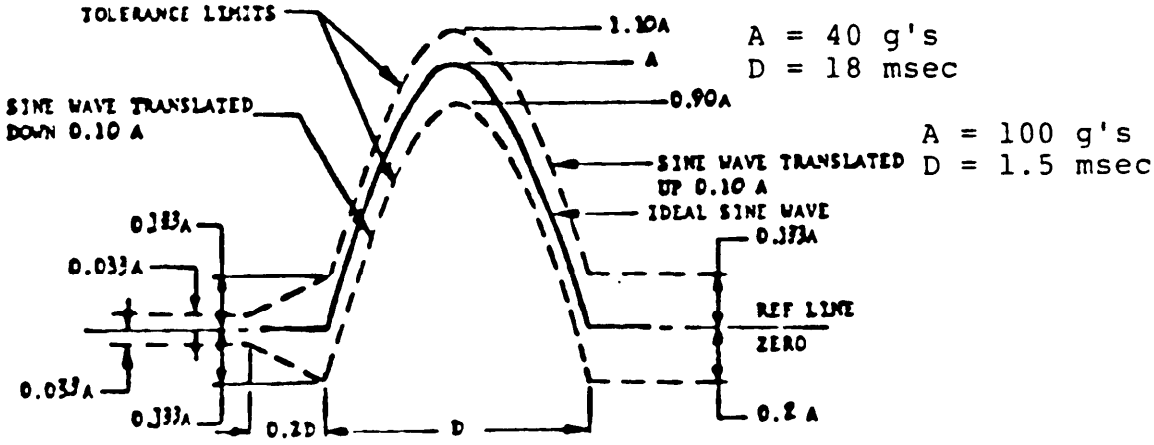


FIGURE 3. Shock Pulse.

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3.4.4 Temperature cycling. The assembly shall be capable of operating as specified herein after withstanding exposure to temperature cycling and soaking between the temperatures of -25°F and 140°F in accordance with Figure 4 (See 4.6.2.5).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in the specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Inspection equipment. Unless otherwise specified in the contract, the contractor shall supply, maintain and calibrate inspection equipment in accordance with the provisions of MIL-STD-45607 and standard commercial practice. Contractor inspection equipment designs shall be submitted for Government approval as specified in the contract (see 6.5).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5 and 6.8).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the quality assurance provisions of MIL-F-13926 and the conditions of 3.1.3.

4.4 First article inspection. The requirements for first article approval and the designation of responsibility for first article inspection to either the Government or the contractor

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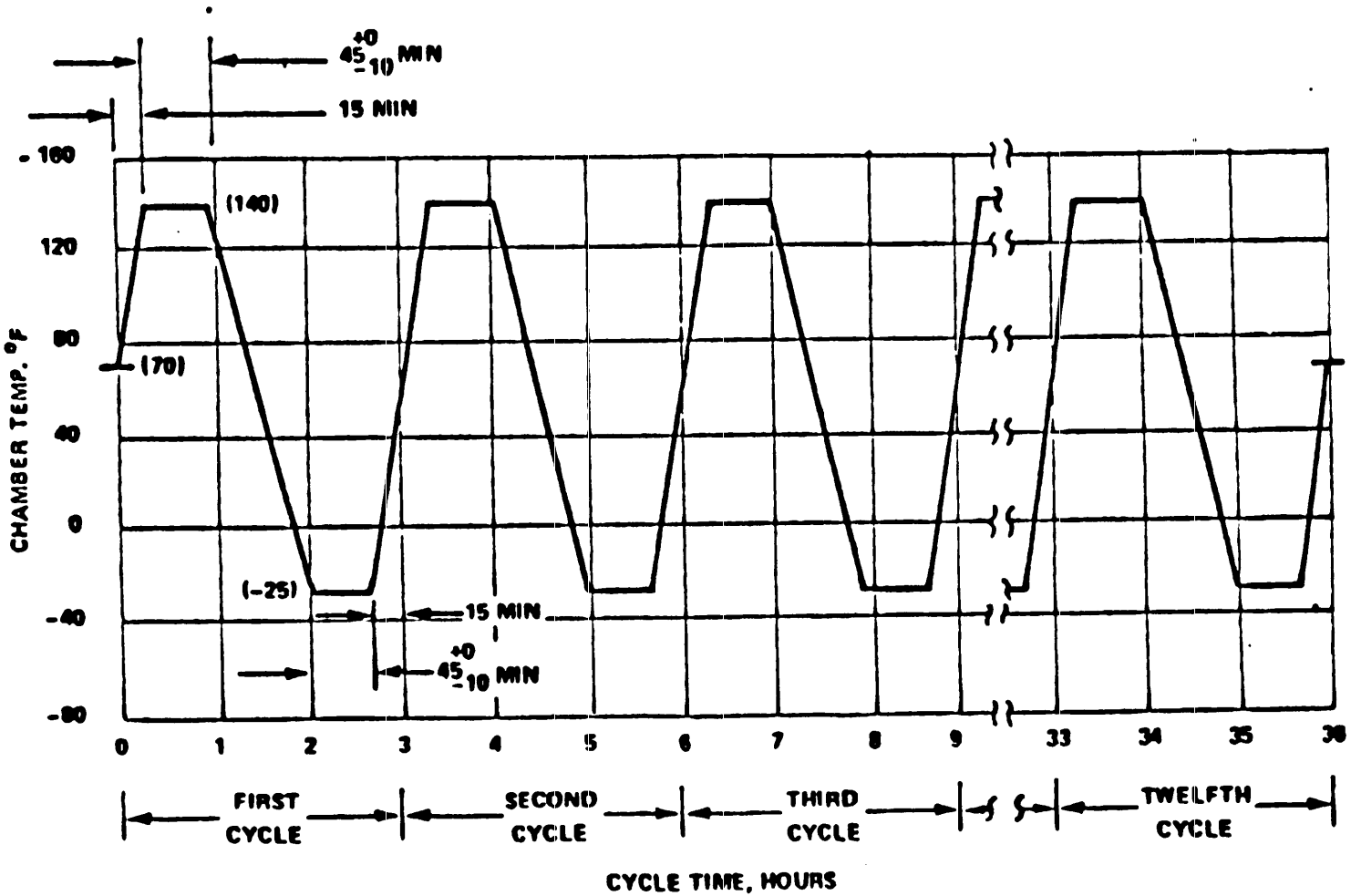


FIGURE 4. Temperature cycling.

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shall be specified in the contract (see 6.3).

4.4.1 Sample. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.4.2. The first article sample shall consist of two assemblies.

4.4.2 Inspection. Two assemblies shall be subjected to all the inspections specified in Tables XIV, XV and XVI. Tests 301 and 302 of Table XVI shall be performed first.

TABLE XIV. Performance tests.

Item	Characteristics	Requirement	Test Procedure
101	$(W-K_WV_W)T_F$ comparator logic	3.3.1	4.6.1.1
102	V_W comparator logic	3.3.2	4.6.1.2
103	Computer A9 card fail logic	3.3.3	4.6.1.3
104	Computer A4 card fail logic	3.3.4	4.6.1.4
105	Computer A5 card fail logic	3.3.5	4.6.1.5
106	Computer A6 card fail logic	3.3.6	4.6.1.6
107	Computer A7 card fail logic	3.3.7	4.6.1.7
108	Computer OK logic	3.3.8	4.6.1.8
109	Dot D_L output logic	3.3.9	4.6.1.9

TABLE XV. Environmental tests.

Item	Characteristics	Requirement	Test Procedure
201	High temperature	3.4.1	4.6.2.1
202	Low temperature	3.4.1	4.6.2.2
203	Vibration	3.4.2	4.6.2.3
204	Shock	3.4.3	4.6.2.4
205	Temperature cycling	3.4.4	4.6.2.5

TABLE XVI. General tests.

Item	Characteristics	Requirement	Test Procedure
301	Fabrication	3.1	Applicable drawings - Visual
302	General specifications	3.1.2	4.6.3 - Visual
303	Packaging inspection	5.1	4.6.4 - Visual

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4.4.3 Failure. Failure of any assembly to meet any requirement shall be cause for refusal to grant first article approval. The Government reserves the right to terminate first article inspection upon any failure of any assembly to comply with any stated requirement.

4.4.4 Responsibility. The contractor, whether or not responsible, shall inspect the sample for conformance to all contractual requirements and shall submit a record of this inspection and certificates of conformance for materials. The Government reserves the right to witness inspections performed by the contractor.

4.5 Quality conformance inspection.

4.5.1 Procurement conditions. The following inspection plans shall apply where the assembly is procured for use as a:

Use	Plan
a. Logistics spare assembly apart from the next higher assembly	A, B and C
b. Component assembly of the next higher assembly	B and C

4.5.2 Inspection plan A.

4.5.2.1 Environmental sample and tests. One assembly, as a control sample, shall be selected at random by the Government representative from each 50 assemblies produced or from each month's production, whichever occurs first. All the tests in Table XV shall be applied separately.

4.5.2.2 Acceptance. When the sample assembly of the produced quantity fails to meet any specified requirement, the lot shall be rejected. Rejected lots shall be resubmitted for reinspection only after all assemblies are re-examined or retested and all defective assemblies are removed or defects corrected.

4.5.3 Inspection plan B.

4.5.3.1 Sample and tests. The sample shall be selected by the Government representative. All the tests in Table XVI and only the test of item 205 in Table XV shall apply. The following sampling plan (except that the visual examination for completeness, improper assembly and evidence of poor workmanship

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per items 301 and 302 shall be performed 100%) shall be utilized:

100%, c=0 for lot sizes of 10 or less
n=10, c=0 for lot sizes of 11 to 150
n=32, c=0 for lot sizes greater than 150

where, n=sample size and c=0 requires zero defects during inspection for acceptance

4.5.3.2 Acceptance. Where any one assembly in the sample fails to meet any specified requirement, the lot shall be rejected. Rejected lots shall be resubmitted as above.

4.5.4 Inspection plan C.

4.5.4.1 Sample and tests. Each assembly in every lot shall be subjected to all the tests in Table XIV and shall be examined visually for completeness, improper assembly and evidence of poor workmanship per items 301 and 302 of Table XVI. The conditions of 3.1.3 apply.

4.5.4.2 Acceptance. Where any one assembly fails to meet any specified requirement, the defective assembly shall be removed from the lot and resubmitted only after all defects have been corrected.

4.6 Methods of inspection.

4.6.1 Performance tests.

4.6.1.1 $(W-K_WV_W)T_F$ comparator logic. Apply the power sources, signal sources, interconnection, and signal inputs per 3.3.1. The output shall be measured and verified to comply with the requirements of 3.3.1.

4.6.1.2 V_W comparator logic. Apply the power sources, signal sources, interconnection, and signal inputs per 3.3.2. The output shall be measured and verified to comply with the requirements of 3.3.2.

4.6.1.3 Computer A9 card fail logic. Apply the power sources, load lamp, interconnection and signal inputs per 3.3.3. Verify load lamp conditions per 3.3.3.

4.6.1.4 Computer A4 card fail logic. Apply the power sources, load lamp, interconnection and signal inputs per 3.3.4. Verify load lamp conditions per 3.3.4.

4.6.1.5 Computer A5 card fail logic. Apply the power sources, load lamp, interconnection and signal inputs per 3.3.5. Verify load lamp conditions per 3.3.5.

4.6.1.6 Computer A6 card fail logic. Apply the power

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sources, load lamp, interconnection and signal inputs per 3.3.6. Verify load lamp conditions per 3.3.6. The output shall be measured and verified to comply with the requirements of 3.3.6.

4.6.1.7 Computer A7 card fail logic. Apply the power sources, load lamp, interconnection and signal inputs per 3.3.7. Verify load lamp conditions per 3.3.7.

4.6.1.8 Computer OK logic. Apply power sources, load lamp, signal sources, interconnection, static input conditions, and signal inputs per 3.3.8. Verify load lamp conditions per 3.3.8.

4.6.1.9 Dot D_L output logic. Apply power sources, interconnection, and signal inputs per 3.3.9. The output shall be measured and verified to comply with the requirements of 3.3.9.

4.6.2 Environmental tests.

4.6.2.1 High temperature. Subject the assembly to the high temperature test specified in MIL-STD-810, Method 501 except storage temperature shall be 160°F for 48 hours. At the conclusion of the storage temperature test, stabilize the assembly at 100°F and subject it to the performance tests of Table XIV. After testing, return the assembly to room temperature and subject it to the performance tests of Table XIV.

4.6.2.2 Low temperature. Subject the assembly to the low temperature test specified in MIL-STD-810, Method 502. Maintain the temperature at -65°F for a period of 12 hours minimum. At the conclusion of the storage temperature test, stabilize the assembly at 40°F and subject it to the performance tests of Table XIV. After testing, return the assembly to room temperature and subject the assembly to the performance tests of Table XIV.

4.6.2.3 Vibration. Vibration A as specified in 4.6.2.3.1 shall be conducted only for first article inspection. Vibration B as specified in 4.6.2.3.2 shall be conducted only for quality conformance inspection.

4.6.2.3.1 Vibration A. Mount the assembly in a suitable vibration fixture and subject the assembly to the vibration test specified in MIL-STD-810, Method 514, General Procedures, and 3.4.2. The vibration amplitude shall have a tolerance of ± 10%. The tolerance of the vibration frequency shall be ± 2%, or 1/2 Hz for frequencies below 25 Hz. Vibration sweeps shall be conducted with the duration of exposure not less than 80 minutes in each of three mutually perpendicular axes. At the conclusion of this test, subject the assembly to the performance tests of Table XIV.

4.6.2.3.2 Vibration B. This test shall be conducted as in 4.6.2.3.1 except that the period of vibration shall be not less than 15 minutes in each axis.

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4.6.2.4 Shock. Mount the assembly in a suitable shock fixture and subject it to the functional shock test for ground equipment specified in MIL-STD-810, Method 516, for a total of three half sine wave shock pulses in both directions along three mutually perpendicular axes (18 shocks). Peak amplitude shall be 40 ± 4 g's with a time duration of 18 ± 3 msec in accordance with Figure 3. At the conclusion of this test, subject the assembly to the performance tests of Table XIV. Repeat this test with peak amplitude of 100.0 ± 10.0 g's with a time duration of 1.5 ± 0.2 msec in accordance with Figure 3.

4.6.2.5 Temperature cycling. With power and loads of Table I applied, subject the assembly to the temperature profile specified in Figure 4. At the conclusion of this test, subject the assembly to the performance tests of Table XIV.

4.6.3 General specifications. Verification shall be provided by examination, visually or through use of standard measuring equipment, in accordance with 3.1.2.

4.6.4 Packaging inspection. The sampling and inspection of the preservation, packaging, packing and container marking shall be in accordance with MIL-STD-2073-1.

5. PACKAGING

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with MIL-STD-2073-1. The level of protection shall be as specified in the procurement document (see 4.6.4 and 6.2d).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The FTS comparator Type III circuit card (A9) assembly is intended for use in the Field Test Set, 11733200, which is used in the field for test of all of the M21 Computer subsystem components.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Inspection provisions for first article (see 6.3).

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d. Applicable levels of preservation, packaging, packing and container marking (see 5.1 and 6.7).

6.3 First article. The contracting officer should provide specific guidance to offerors for the submission of a first article sample to be tested as specified in 4.4.2. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Supersession. Previous revisions of MIL-C-48485(AR) were for the US Army Armament, Research, Development and Engineering Center's internal use only, and were not forwarded to the Naval Publications and Form Center for publication.

6.5 Submission of contractor inspection equipment designs for approval. Submit copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: AMSTA-AR-QAC-F, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.6 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament, Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal. Technical data originally prepared by these actvities is now under the cognizance of ARDEC.

6.7 Levels of packaging. The following definitions describe the three levels of packaging in accordance with MIL-STD-2073-1.

a. Level A - Adequate protection for indeterminate storage and worldwide distribution.

b. Level B - Adequate protection for known shipment conditions and storage conditions normally not exceeding one year.

c. Level C - Adequate protection for shipment to a known site for immediate use.

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6.8 Alternative quality conformance provisions. Alternative quality conformance procedures, methods, or equivalent, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equivalent, the contractor shall describe them in a written proposal submitted to the Government for evaluation. When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality assurance provision(s) herein. In cases of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable. Unless otherwise specified in the contract, proposed alternative quality conformance provisions will be submitted by the contractor for evaluation by the technical activity.

6.9 Subject term (key word) listing.

Computer, M21
Fire Control
Tank, M60A3

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1240-0012)

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-48485C (AR)	950623

3. DOCUMENT TITLE: CIRCUIT CARD ASSEMBLY - COMPARATOR, TYPE 3

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTED BY

6a. NAME (Last, First, Middle Initial)	6b. ORGANIZATION
6c. ADDRESS (Include Zip Code)	6d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

8a. NAME	8b. TELEPHONE (Include Area Code)
U.S. ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial: 201-724-6629 (2) AUTOVON: DSN-880-6629
8c. ADDRESS (Include Zip Code) ATTN: AMSTA-AR-EDE-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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